

Jasmine Latendresse

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Research Interests

Empirical software engineering; software ecosystems and dependency management; AI-assisted software development; software engineering for AI.

Education

Concordia University, Montréal, QC

PhD, Software Engineering (fast-track), 2021–2026

GPA: 4.25/4.30

Supervisor: Prof. Emad Shihab

Dissertation: *Understanding and Supporting Software Dependency Management Across the Lifecycle*

Defense outcome: Outstanding

Concordia University, Montréal, QC

MSc, Software Engineering, 2020–2021

Fast-tracked to PhD program

GPA: 4.23/4.30

Concordia University, Montréal, QC

BEng, Software Engineering, 2015–2020

GPA: 3.29/4.30

Peer-reviewed Publications

Journal Articles

Jasmine Latendresse, Samuel Abedu, Ahmad Abdellatif, and Emad Shihab. “An Exploratory Study on Machine Learning Model Management.” *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2023.

Conference Proceedings

Jasmine Latendresse, SayedHassan Khatoonabadi, and Emad Shihab. “The Software Librarian: Python Package Insights for Copilot.” In *Proceedings of the IEEE/ACM 47th International Conference on Software Engineering: Companion Proceedings (ICSE Companion 2025)*, April 2025.

Thomas Le Tourneau, **Jasmine Latendresse**, Ahmad Abdellatif, and Emad Shihab. “Code Mapper: Mapping the Global Contributions of OSS.” In *Proceedings of the 46th International Conference on Software Engineering (ICSE 2024)*, April 2024.

Jasmine Latendresse, Suhaib Mujahid, Diego Elias Costa, and Emad Shihab. “Not All Dependencies Are Equal: An Empirical Study on Production Dependencies in NPM.” In *Proceedings of the 37th International Conference on Automated Software Engineering (ASE 2022)*, October 2022.

Jasmine Latendresse, Rabe Abdalkareem, Diego Elias Costa, and Emad Shihab. “How Effective Is Continuous Integration in Indicating Single-Statement Bugs?” In *Proceedings of the 18th International Conference on Mining Software Repositories (MSR 2021)*, May 2021.

Manuscripts Under Review

Jasmine Latendresse and Emad Shihab. “Evaluating Large Language Models for Dependency Maintenance Patch Generation.” Manuscript submitted to *ACM Transactions on Software Engineering and Methodology*, 2026.

Jasmine Latendresse, SayedHassan Khatoonabadi, and Emad Shihab. “How Robust Are LLM-Generated Library Imports? An Empirical Study Using Stack Overflow.” Revised manuscript under review at *IEEE Transactions on Software Engineering*, 2026.

Research & Professional Experience

AI Researcher, National Bank of Canada (NBC), Montréal, QC — April 2025–Present

- Designed an AI-driven approach for assessing change-request risk in IT service management.
- Built and evaluated ML and LLM models for risk classification and generation of decision-support explanations.
- Analyzed large-scale change and incident datasets to identify actionable risk signals and inform feature engineering.
- Contributed to the design and development of a prototype decision-support system for early risk detection.

AI Scientist, Ministère de la Cybersécurité et du Numérique (MCN), Gouvernement du Québec, Montréal, QC — 2023–2025

- Led a team of AI engineers in the design, deployment, and evaluation of an LLM-based writing assistant for government employees.

- Deployed and optimized open-source large language models locally using tools such as llama.cpp to satisfy privacy and security requirements.
- Conducted extensive benchmarking, reliability testing, and model evaluation to select and optimize models for French- and English-language use cases.
- Contributed to the development of secure and privacy-conscious approaches for deploying generative AI within government environments.

Research Assistant, Data-driven Analysis of Software (DAS) Lab, Concordia University, Montréal, QC — May 2020–2026

- Conducted empirical studies on continuous integration, dependency management, ML model management, and AI-based software systems.
 - Designed and executed large-scale empirical studies using software repository and ecosystem data.
 - Mined and analyzed data from sources including GitHub and npm using Python, statistical analysis, and data-processing workflows.
 - Developed research prototypes and experimental pipelines for evaluating software engineering tools and AI-based systems.
 - Collaborated with researchers and students on study design, implementation, analysis, and scientific writing.
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Research Mentoring & Supervision

- Mentored **three student researchers** through structured research projects, providing guidance on research questions, empirical methodology, implementation, data analysis, and scientific writing.
- Mentoring activities contributed to peer-reviewed publications including “**The Software Librarian: Python Package Insights for Copilot**” (ICSE Companion 2025) and “**Code Mapper: Mapping the Global Contributions of OSS**” (ICSE 2024).
- Currently supervise and support research activities conducted within the Concordia University–National Bank of Canada research partnership.
- Regularly mentor undergraduate and graduate researchers in empirical software engineering methods, research tooling, experimental design, and academic writing.

Teaching Experience

Teaching Assistant, Engineering AI-based Software Systems, Department of Computer Science & Software Engineering, Concordia University — 2023-2026

- Led tutorials on ML/LLM-based system design, evaluation, and deployment.

- Supported graduate students with project design, implementation, and empirical evaluation.
 - Developed lecture materials and delivered lectures on course topics related to AI-based software systems
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Academic Service

Junior Program Committee Member, MSR Research Track — 2023–2025

Reviewer, ACM TOSEM — 2025–2026

Research Training

NSERC CREATE Program — Program Graduate

- Completed interdisciplinary research training through the Natural Sciences and Engineering Research Council of Canada CREATE program.
 - Participated in research, professional-development, and collaborative training activities alongside graduate researchers.
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Awards & Honours

- Graduate Students' Association (GSA) Conference Funding, Concordia University, 2023.
 - Concordia Golden Key Scholarship, Golden Key International Honour Society, 2022.
 - Concordia University Conference and Exposition Award, Concordia University, 2022.
 - Concordia University Retired Faculty and Staff Graduate Award, Concordia University, 2021.
 - Doctoral Fellowship, Concordia University, 2021.
 - Merit Scholarship, Concordia University, 2021.
 - Instructional Technology Award, Concordia University, 2020.
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Languages

French: Native

English: Full professional proficiency